

CH10.2 三角函數圖形與疊合公式

1.(A)(C) 2.(B)(C) 3. $a > b > d > c$ 4.(A)(E) 5.5

6.4 7.7 8. $-\sqrt{3}$ 9. $e = d > c > b > a$ 10.0 或 $\frac{\pi}{4}$ 解析

1. (A) $\circ: -1 \leq \cos\left(2x + \frac{\pi}{3}\right) \leq 1 \Rightarrow -6 \leq y \leq 6$

(B) X : 週期為 $\frac{2\pi}{2} = \pi$

(C) $\because 2 \times \frac{\pi}{3} + \frac{\pi}{3} = \pi$

(D) $x: y = 6\cos\left(2 \times \left(-\frac{\pi}{6}\right) + \frac{\pi}{3}\right) = 6\cos 0 = 6 \neq -6$

(E) $\times$: 應為向左平移 $\frac{\pi}{6}$

故選(A) (C)

2. $f(x) = 2\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right)$

$$= 2\left(\sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4}\right)$$

$$= 2\sin\left(x + \frac{\pi}{4}\right)$$

(A) $\times$: 應為 $-2 \leq f(x) \leq 2$

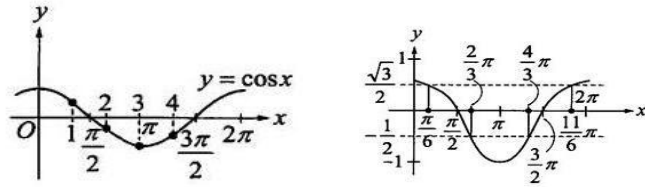
(B) $\circ: f(0) = 2\sin \frac{\pi}{4} = \sqrt{2}$, 交點為 $(0, \sqrt{2})$

(C) $\circ: f\left(\frac{\pi}{4}\right) = 2\sin\left(\frac{\pi}{4} + \frac{\pi}{4}\right) = 2$

(D) $\times$: 應為 2π ; (E) $\times: \frac{3\pi}{4} + \frac{\pi}{4} = \pi$

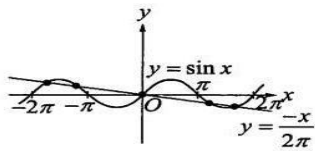
故選(B)(C)

3. 描繪 $y = \cos x$ 的圖形如右: $\frac{\pi}{2} \approx 1.57, \pi \approx 3.14, \frac{3\pi}{2} \approx 4.71$ 由右圖中四點位置的高低, 得 $\cos 1 > \cos 2 > \cos 4 > \cos 3$ 故 $a > b > d > c$



4. 由右圖知所求為 $\frac{\pi}{6} < x < \frac{2\pi}{3}$ 或 $\frac{4\pi}{3} < x < \frac{11\pi}{6}$

故選(A)

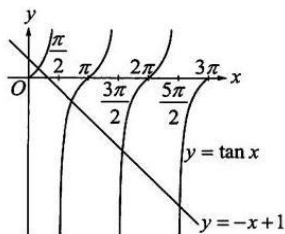


$$5. x + 2\pi \sin x = 0$$

$$\Leftrightarrow \sin x = \frac{-x}{2\pi}$$

有 5 個交點, 故有 5 個實根

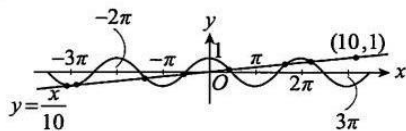
6.



共 4 個交點

$$7. 10\cos x = x \Leftrightarrow \cos x = \frac{x}{10}$$

$$\text{考慮 } \begin{cases} y = \cos x \\ y = \frac{x}{10} \end{cases}$$



兩圖形的交點個數即為方程式的實數解，

由右圖可知共有 7 個交點

$$\Rightarrow \cos x = \frac{x}{10} \text{ 有 } 7 \text{ 個實根}$$

$$8. y = \left(\sin x \cos \frac{\pi}{6} - \cos x \sin \frac{\pi}{6} \right) + 2\cos x$$

$$= \frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x + 2\cos x = \frac{\sqrt{3}}{2} \sin x + \frac{3}{2} \cos x$$

$$= \sqrt{3} \left(\sin x \cdot \frac{1}{2} + \cos x \cdot \frac{\sqrt{3}}{2} \right)$$

$$= \sqrt{3} \left(\sin x \cos \frac{\pi}{3} + \cos x \cdot \sin \frac{\pi}{3} \right) = \sqrt{3} \sin \left(x + \frac{\pi}{3} \right)$$

$$\because 0 \leq x \leq \frac{3\pi}{2} \therefore \frac{\pi}{3} \leq x + \frac{\pi}{3} \leq \frac{11\pi}{6}$$

$$\text{當 } x + \frac{\pi}{3} = \frac{3\pi}{2}$$

$$\Rightarrow x = \frac{7\pi}{6} \text{ 時，有最小值為 } \sqrt{3} \sin \frac{3\pi}{2} = -\sqrt{3}$$

$$9. a = \sin 10^\circ + \cos 10^\circ = \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin 10^\circ + \frac{1}{\sqrt{2}} \cos 10^\circ \right) = \sqrt{2} \sin(10^\circ + 45^\circ) = \sqrt{2} \sin 55^\circ$$

$$\text{同理 } b = \sqrt{2} \sin 65^\circ, c = \sqrt{2} \sin 75^\circ, d = \sqrt{2} \sin 85^\circ,$$

$$e = \sqrt{2}\sin 95^\circ$$

$$\Rightarrow e = d > c > b > a$$

$$10. \quad 1 = \cos 2x + \sin 2x = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 2x + \frac{1}{\sqrt{2}} \sin 2x \right)$$

$$= \sqrt{2} \sin \left(2x + \frac{\pi}{4} \right) \Rightarrow \sin \left(2x + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$$

$$\text{因為 } 0 \leq x < \pi, \text{ 所以 } \frac{\pi}{4} \leq 2x + \frac{\pi}{4} < \frac{9\pi}{4}$$

$$\Rightarrow 2x + \frac{\pi}{4} = \frac{\pi}{4} \text{ 或 } \frac{3\pi}{4} \Rightarrow x = 0 \text{ 或 } \frac{\pi}{4}$$